LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	16
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Secondary/Auxiliary flight controls, thrust reversing/braking
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Secondary/Auxiliary flight controls, thrust reversing/braking
5. Specific Objectives	6.1.2.1 Trainee will be able to describe various types of aerofoil sections, conditions of performance, theories, and concept related with performance

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of L & Ds diagrams for Spoilers, Flaps, and Thrust Reverser mechanisms; Instructor supervision on safety protocols regarding hydraulic systems and brake actuation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary flight controls?
(b) Define the function of ailerons.
(c) List the three main types of secondary flight controls.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and purpose of secondary controls • Types of flaps and their aerodynamic effects • Spoiler operation and ground effect • Trim systems and balance mechanisms • Hydraulic actuation principles	L & Ds	PPT, IB	15 min
(b)	• Thrust reverser mechanical systems • Brake actuation and anti-skid systems • Integration of reverse thrust with braking • Safety interlocks and ground handling • Maintenance checks for reverse thrust units	L & Ds	PPT, PDF	20 min
(c)	• Performance impact of flap deployment • Braking efficiency calculations • Emergency braking procedures • System failure scenarios and recovery • Operational limits for secondary controls	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how spoilers assist in roll control.
(b) Describe the sequence of thrust reverser deployment.
(c) Identify the role of anti-skid systems during landing.

12. Home Assignment and Follow Up	Prepare a brief report comparing the aerodynamic benefits of leading-edge flaps versus trailing-edge flaps, referencing LP 02 concepts.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Flight control surface balancing	Refer to the master syllabus reference list	LP 17

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Senior Instructor, P No. 12345	Senior Officer, Chief Instructor, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min